Weichen Wang — Curriculum Vitae

Contact Information

Department of Physics,
University of Milano-Bicocca
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Education

Johns Hopkins University, Baltimore MD, United States — 9/2016 - 12/2022
Ph. D. in Astrophysics, Department of Physics and Astronomy
Thesis Advisor: Susan Kassin (STScI); JHU Advisor: Timothy Heckman
Thesis title: *Observational studies of the relation between galactic winds and star formation at half the age of the Universe*

Tsinghua University, Beijing, China — 8/2012 - 7/2016
B. Sc. in Physics, Department of Physics
Thesis Advisor: Shude Mao

Research Positions

University of Milano-Bicocca, Milan MI, Italy — 12/2022 - now
Postdoc researcher, Department of Physics
Advisor: Sebastiano Cantalupo

University of California, Santa Cruz CA, United States — 2020; 7-9/2015
Visiting student, Department of Astronomy
Advisors: Sandra Faber, David Koo

Research Interests

Galaxy Formation in the Early Cosmic Epochs
Galaxy Formation in Extreme Large-scale Environments
Spectroscopic Studies of Galaxies and Their Circumgalactic Medium

Publications

Four first-author papers published in Nature Astronomy, ApJ, and MNRAS.
A total of 17 publications with 1000+ citations in the past 3 years; h-index: 15
See the attached publication list or this ADS link for full publication records.

Awards

Giovani Talenti (Young Talents Award)
by U. Milano-Bicocca & Accademia Nazionale dei Lincei of Italy, 2025

The International Astronomical Union Travel Grant, 2019

First-year student award, Johns Hopkins Physics & Astronomy Deptartment, 2016

National Astronomical Observatory of China Scholarship, 2016

Award of excellence by Tsinghua University: 清华学堂计划, 2013-2016

Media

Press releases by the European Space Agency, Nature Portfolio, U. Milano-Bicocca, Caltech, Swinburne U. (reporting W. Wang+ 2025):
"Big Wheel", a surprisingly large disk galaxy discovered in the early universe.

News coverage by Scientific American, The Independent, Newsweek, Daily Mail, Il Giornale: *JWST Spots Giant Spiral Galaxy Shockingly Early in Cosmic History.*

News coverage by Phys.org (reporting W. Wang+ 2026):
'Red Potato' galaxy discovered by astronomers.

TALKS

Invited Conference Talks:
The Controversial Fate of High-z Galaxy Protoclusters, Spain, 2025
Galaxies and Diffuse Gas in Large-Scale Overdense Environments, Italy, 2024

Department Seminars:
Tsinghua University/Peking University/NAOC, China, 2023/2024
University of Missouri, U.S., 2020

Recent Contributed Talks (Selected):
The Baryon Cycle from Reionization to the Cosmic Noon, Chile, 2025
Tracing Cosmic Evolution with Galaxy Clusters, Italy, 2025
Observing and Simulating Galaxy Evolution in the Era of JWST, Switzerland, 2024
What Matter(s) Around Galaxies 2024, Italy, 2024
Steward/NOIRLab Galaxy Group Lunch Talk, University of Arizona, U.S., 2021
Massively Parallel Large Area Spectroscopy from Space, Portugal, 2021

SERVICE

SOC/LOC member, Conference What Matter(s) Around Galaxies, 2024
Referee for The Astrophysical Journal, Astronomy and Astrophysics

TECHNICAL EXPERTISE

JWST Spectroscopy and Imaging:
Serving as the deputy lead for JWST program GO1835, a spectroscopic survey of more than 50 galaxies in two $z \sim 3$ QSO fields; responsible for the planning, implementation, and analysis of the NIRSpec and NIRCam observations;

Keck spectroscopy: Reducing, organizing, and analyzing the spectra of $\simeq$200 z=1 galaxies as the member of a Keck/DEIMOS program HALO7D (P.I.: GuhaThakurta).

OBSERVATION PROGRAMS (SELECTED)

Summary: 100+ hours on JWST and 120+ hours on VLT & ALMA. Programs joined as P.I. and core member are listed and marked with [P] and [C], respectively.

[C] ALMA Cycle-12 Program (P.I.: A. Pensabene; 27 hours):
The Big Wheel Galaxy: a unique laboratory to study the physics of early disk formation in overdense environments, 2025-2026

[C] JWST Cycle-4 Program (P.I.: B. Wang; 20 hours):
Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking object by hunting for stellar absorption features, 2025-2026

[C] VLT P112 Program (P.I.: S. Cantalupo; 84 hours): *Connecting the dots with MUSE: the Cosmic Web in emission around a massive structure at z=3,* 2024-2025

JWST Cycle-2 Program (P.I.: Kassin; 68 hr): *Galaxy angular momentum alignment with filaments at $z \sim 3$: The effect of large scale structure on galaxies,* 2024

[C] JWST Cycle-1 Program (P.I.: S. Kassin; 74 hours): *A Pathfinder for JWST Spectroscopy: Deep High Spectral Resolution Maps of Galaxies over $1<z<6$,*, 2023

[C] JWST Cycle-1 Program (joined in 2022 with major contributions; P.I.: S. Cantalupo; 24 hours): *Unraveling the Knots of Gaseous Cosmic Web Filaments at $z \sim 3$ through H-alpha Emission Observations,* 2023

ALMA Cycle-8 Program (P.I.: R. Simons): *CO Kinematics at Cosmic Noon: Timing the Redistribution of Metals Around Galaxies,* 23.1 hours, 2022

[P] ALMA Cycle-7 Program (P.I.; 15 hours): *Does molecular gas follow the motion of ionized gas inside typical high-redshift star-forming galaxies?* Observations not completed due to weather and the impact of COVID-19 in Chile, 2021

Mentorship

Xiaohan Wang, Visiting Ph.D student of Tsinghua University, 2025
JWST NIRSpec studies of galaxy metallicity in the high-redshift cosmic web
Resulting in one research paper completed & submitted to A&A.

M. Francesca Uboldi, Bicocca undergraduate in physics major, 2024 (thesis project)
Relations between galaxy colors and morphology with JWST medium-band imaging
Now pursuing Masters in Astronomy degree in Bicocca.

Ying Qin, Johns Hopkins undergraduate in physics, 2021-2023
Studying the Mg II emission and ionizing photons from low-mass galaxies at $z \sim 1$
Resulting in two conference poster presentations, including one at the AAS Annual Meeting; Now pursuing a career as lawyer in Washington D.C.

Teaching Experience

Volunteer Teaching Assistant, Laboratory of Data Analysis in Astrophysics
University of Milano-Bicocca, Spring 2023

Teaching Assistant, General Physics I for Biological Science Majors
Johns Hopkins University, Fall 2016

Teaching Assistant, General Physics Laboratory
Johns Hopkins University, Fall 2016

Outreach Activities

Science outreach day at University of Milano-Bicocca, 2024

A community event open to students from local primary/middle schools in Milan; co-designing the exhibition about astrophysical science with JWST

Member of the Astro Scholars program, Johns Hopkins University, 2021-2022

An annual program about astrophysics and coding for college students from underrepresented backgrounds; serving as a core member of the hiring & education team

Member of the Physics and Astronomy Graduate Students Outreach Team (PAGS), Johns Hopkins University, 2017-2019

Supporting visits of students from Baltimore local primary/middle schools around once per semester and teaching fundamental physics with educational demos

Volunteer teacher at the Pengzhai Primary School, Guizhou, China, Summer 2013

Teaching multiple STEM-related courses for Grade 3-6; the school, with very limited resources, is located in one of the least developed areas of the country.

Weichen Wang — Publication List

Overview

Four first-author papers published in Nature Astronomy, ApJ, and MNRAS
A total of 17 publications with 1000+ citations in the past 3 years; h-index: 15
Full records are also available on the NASA ADS via this link.

Research topic categories marked below:
C1: Early-epoch galaxy formation in extreme cosmic environments; **C2:** Interplay between galaxies and the cosmic gas; **C3:** Galaxy star formation and dust.

First-Author Papers

[5,C1] **W. Wang**, S. Cantalupo, M. Galbiati, et al. arXiv: 2601.20473, submitted to A&A (2026): *A Quiescent Galaxy in a Gas-Rich Cosmic Web Node at z∼3*

[4,C1] **W. Wang**, S. Cantalupo, A. Pensabene, et al. Nature Astron., 9, 710 (2025): *A Giant Disk Galaxy Two Billion Years After The Big Bang* (Editorial Highlight)

[3,C2] **W. Wang,** S. A. Kassin, S. M. Faber, D. C. Koo, et al., ApJ, 930, 146 (2022): *The Baltimore Oriole's Nest: Cool Winds from the Inner and Outer Parts of a Star-Forming Galaxy at* $z = 1.3$

[2,C3] **W. Wang,** S. A. Kassin, C. Pacifici, et al., ApJ, 869, 161 (2018): *Galaxy Inclination and the IRX-β Relation: Effects on UV Star Formation Rate Measurements at Intermediate to High Redshifts*

[1,C3] **W. Wang,** S. M. Faber, F.-S. Liu, et al., MNRAS, 469, 4063 (2017): *UVI colour gradients of 0.4<z<1.4 star-forming main-sequence galaxies in CANDELS: dust extinction and star formation profiles*

Publications [1-3] are all referenced by recent review articles on the Annual Review of Astronomy and Astrophysics (ARA&A).

Papers with Leading Contributions

[2,C1] X. Wang*, S. Cantalupo, **W. Wang**, et al. submitted to A&A (2025): *Metal enrichment of galaxies in a massive node of the Cosmic Web at* $z \sim 3$

[1,C1] A. Pensabene, S. Cantalupo, **W. Wang**, et al. A&A, 701, A120 (2025): *ALMA survey of a massive node of the Cosmic Web at z∼3: II. A dynamically cold disk galaxy in the proximity of a hyperluminous quasar*

* paper led by PhD student co-mentored

Co-Author Publications

[22] M. Galbiati, et al., A&A, 696, A95 (2025): *Connecting the growth of galaxies to the large-scale environment in a massive node of the Cosmic Web at z∼3*

[21] A. de la Vega, et al., ApJ, 980, 168 (2025): *Improved SED-fitting Assumptions Result in Inside-out Quenching at z∼0.5 and Quenching at All Radii Simultaneously at z∼1*

[20] A. Travascio, et al., A&A, 694, A165 (2025): *X-ray view of a massive node of the Cosmic Web at z∼3: I. An exceptional overdensity of rapidly accreting supermassive black holes*

[19] Y. Liang, et al., ApJ, 986, 60 (2025): *Cosmic Himalayas: The Highest Quasar Density Peak Identified in a 10,000 deg2 Sky with Spatial Discrepancies between*

Galaxies, Quasars, and IGM H I

[18] S. L. Finkelstein, et al., ApJL, 983, L4 (2025): *The Cosmic Evolution Early Release Science Survey (CEERS)*

[17] J. Ren, et al., ApJ, 982, 200 (2025): *The Evolution of the Size and Merger Fraction of Submillimeter Galaxies across 1<z<6 as Observed by JWST*

[16] A. Pensabene, et al., A&A, 684, A119 (2024): *ALMA survey of a massive node of the Cosmic Web at z∼3. I. Discovery of a large overdensity of CO emitters*

[15] M. Li, et al., ApJS, 275, 27 (2024): *MAMMOTH-Subaru. II. Diverse Populations of Circumgalactic Lyα Nebulae at Cosmic Noon*

[14] X. Wang, et al., MNRAS, 533, 2026 (2024): *SDSS-IV MaNGA: stellar rotational support in disc galaxies versus central surface density and stellar population age*

[13] S. de Beer, et al., MNRAS, 526, 1850 (2023): *Resolving the physics of quasar Lyα nebulae (RePhyNe): I. Constraining quasar host halo masses through circumgalactic medium kinematics*

[12] P. G. Pérez-González, et al., ApJL, 946, L16 (2023): *CEERS Key Paper. IV. A Triality in the Nature of HST-dark Galaxies*

[11] Y. Guo, et al., ApJL, 945, L10 (2023): *First Look at z>1 Bars in the Rest-frame Near-infrared with JWST Early CEERS Imaging*

[10] C. Pacifici, et al., ApJ, 944, 141 (2023): *The Art of Measuring Physical Parameters in Galaxies: A Critical Assessment of Spectral Energy Distribution Fitting Techniques*

[9] J. A. Zavala, et al., ApJL, 943, L9 (2023): *Dusty Starbursts Masquerading as Ultra-high Redshift Galaxies in JWST CEERS Observations*

[8] S. L. Finkelstein, et al., ApJL, 940, L55 (2022): *A Long Time Ago in a Galaxy Far, Far Away: A Candidate z∼12 Galaxy in Early JWST CEERS Imaging*

[7] J. Pharo, et al., ApJS, 261, 12 (2022): *The Dwarf Galaxy Population at z∼0.7: A Catalog of Emission Lines and Redshifts from Deep Keck Observations*

[6] R. C. Simons, et al., ApJ, 874, 59 (2019): *Distinguishing Mergers and Disks in High-redshift Observations of Galaxy Kinematics*

[5] F. S. Liu, et al., ApJ, 860, 60 (2018): *On the Transition of the Galaxy Quenching Mode at 0.5<z<1 in CANDELS*

[4] D. Jiang, et al., ApJ, 854, 70 (2018): *The Isophotal Structure of Star-forming Galaxies at 0.5<z<1.8 in CANDELS: Implications for the Evolution of Galaxy Structure*

[3] Y. Guo, et al., ApJ, 853, 108 (2018): *Clumpy Galaxies in CANDELS. II. Physical Properties of UV-bright Clumps at 0.5<z<3*

[2] F. S. Liu, et al., ApJL, 844, L2 (2017): *The Origins of UV-optical Color Gradients in Star-forming Galaxies at z∼2: Predominant Dust Gradients but Negligible sSFR Gradients*

[1] F. S. Liu, et al., ApJL, 822, L25 (2016): *The UV-Optical Color Gradients in Star-forming Galaxies at 0.5<z<1.5: Origins and Link to Galaxy Assembly*